

Diagnostic Evaluation Report

1. Framework Information

- **Framework:** X-RAG Diagnostic Framework (v1.0)
- **Trace ID:** `d7bda32a-7dc4-4f05-b55f-2a96fcf0f510`
- **Timestamp:** 2026-09-05T15:21:12.363293Z
- **Status:** COMPLETED

2. Executive Summary

- **Question:** In [2022] 15 S.C.R. 104, Kerala General Sales Tax Act, 1963 – s.10 – Sick Industrial Companies (Special Provisions) Act, 1985 – s.3(1)(b) –
- **Answer:** I do not have enough information to answer this.
- **Health Score:** 0.07
- **Primary Issue:** GROUNDING_FAILURE
- **Claims Supported:** 0 / 1
- **Summary:** Analysis completed.

3. Pipeline Overview

Stage	Status	Confidence	Observation
CORPUS	UNKNOWN	0.5	Corpus relevance is inconclusive.
RETRIEVER	UNKNOWN	0.5	Retrieval evidence does not definitively confirm or refute pipeline success.
GENERATOR	PASS	0.95	All generated claims are fully supported by evidence.
GROUNDING	FAIL	0.95	1 claim(s) are actively contradicted by retrieved evidence.

Efficiency: 0/6 chunks used (0%)

4. Evaluation Metrics

- **Retrieved Chunks:** 6
- **Verified Claims:** 1 (Supported: 0, Partial: 0, Unsupported: 0, Contradicted: 1)
- **Grounding Score:** 0.00
- **Evidence Coverage:** 0.17
- **Average Entailment:** 0.06
- **Latencies:** Retrieval (2348.1ms) | Generation (7750.1ms) | Verification (3977.4ms)

5. RAGAS Metrics

- **Faithfulness:** 0.0000
- **Answer Relevancy:** 0.3285
- **Context Precision:** 0.0000
- **Context Relevancy:** 0.0000
- **Context Recall:** N/A
- **Answer Similarity:** N/A
- **Answer Correctness:** N/A

Answer Correctness (Claim Recall)

Not computed for this trace.

6. Evidence Analysis

- **[748b9288-e841-47e6-814d-a7db13911ebc]** (CONTRADICTED): I do not have enough information to answer this.
- *Evidence (Chunk 5b3fb972-5d61-4b66-864e-9d0edf0a9e7b):* of any property, or to execute any judicial process, or to administer the law, or to interpret, or to preserve order in the Court, and every person specially authorised by a Court to perform any of such duties;
(d) every assessor or member of a panchayat assisting a Court or public servant;
(e) every arbitrator or other person to whom any cause or matter has been referred for decision or report by any Court, or by any other competent public authority;
(f) every person who holds any office by virtue of which he is empowered to place or keep any person in confinement;
(g) every officer of the Government whose duty it is, as such officer, to prevent offences, to give information of offences, to bring offenders to justice, or to protect the public health, safety or convenience;
(h) every officer whose duty it is, as such officer, to take, receive, keep or

expend any property on behalf of the Government, or to make any survey, assessment or contract on behalf of the Government, or to execute any revenue-process, or to investigate, or to report, on any matter affecting the pecuniary interests of the Government, or to make, authenticate or keep any document relating to the pecuniary interests of the Government, or to prevent the infraction of any law for the protection of the pecuniary interests of the Government;

(i) every officer whose duty it is, as such officer, to take, receive, keep or expend any property, to make any survey or assessment or to levy any rate or tax for any secular common purpose of any village, town or district, or to make, authenticate or keep any document for the ascertaining of the rights of the people of any village, town or district;

(j) every person who holds any office by virtue of which he is empowered to prepare, publish, maintain or revise an electoral roll or to conduct an election or part of an election;

(k) every person—

(i) in the service or pay of the Government or remunerated by fees or commission for the performance of any public duty by the Government;

(ii) in the service or pay of a local authority as defined in clause (31) of section 3 of the General Clauses Act,

7. Root Cause Analysis

- **Primary Cause:** GROUNDING_FAILURE

- **Confidence:** 0.95

- **Reasoning:**

- Starting root cause analysis for trace d7bda32a-7dc4-4f05-b55f-2a96fcf0f510.
- Skipping CORPUS: status is UNKNOWN (insufficient evidence).
- Skipping RETRIEVER: status is UNKNOWN (insufficient evidence).
- Skipping CHUNKING: no observable state in matrix.
- GENERATOR passed (confidence=0.95).
- Failure at GROUNDING: GROUNDING_FAILURE (confidence=0.95). Observation: 1 claim(s) are actively contradicted by retrieval.
- Primary root cause: GROUNDING_FAILURE at GROUNDING (earliest upstream failure, confidence=0.95).
- Analysis complete. Primary cause: GROUNDING_FAILURE.

8. Corrective Action Plan

- **[EXPERIMENTAL] Require Explicit Citations**

- Description: Force the LLM to append [Chunk-ID] to every sentence it generates.
- Expected Improvement: Self-correction during generation ensures every fact is linked to evidence. (Metric: 100% of claims supported)
- **[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff**
- Description: Retrieval efficiency is low (0%); consider reducing top-k or tightening the reranker score cutoff.
- Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utilization)

9. Evidence Traceability

- **Primary Cause:** GROUNDING_FAILURE

- **Derived From:** RootCauseAnalysis

- **Pipeline Stage:** GROUNDING

- **Supporting Claims:** 748b9288-e841-47e6-814d-a7db13911ebc

- **Supporting Chunks:** 5b3fb972-5d61-4b66-864e-9d0edf0a9e7b

10. Overall Assessment

- **Major Strength:** GENERATOR is healthy (confidence=95%): All generated claims are fully supported by evidence.
- **Major Weakness:** GROUNDING_FAILURE
- **Next Priority:** Require Explicit Citations
- **Recommendation:** Pipeline requires architectural adjustments across multiple stages.

11. Framework Footer

- **Generated By:** X-RAG Diagnostic Framework

- **Artifact Version:** 1.0

APPENDIX

Artifacts Generated

✓ RAGTrace

artifacts/rag_traces/d7bda32a-7dc4-4f05-b55f-2a96fcf0f510.json

✓ ClaimSet

artifacts/claim_sets/d7bda32a-7dc4-4f05-b55f-2a96fcf0f510.json

✓ VerificationResults

artifacts/verification/d7bda32a-7dc4-4f05-b55f-2a96fcf0f510.json

✓ PipelineStateMatrix

artifacts/pipeline_state_matrix/d7bda32a-7dc4-4f05-b55f-2a96fcf0f510.json

✓ RootCauseAnalysis

artifacts/root_cause_analysis/d7bda32a-7dc4-4f05-b55f-2a96fcf0f510.json

✓ CorrectiveActionPlan

artifacts/corrective_action_plan/d7bda32a-7dc4-4f05-b55f-2a96fcf0f510.json

✓ DiagnosticEvaluationReport

artifacts/reports/d7bda32a-7dc4-4f05-b55f-2a96fcf0f510.json